

LESSON 3.9

SOLVING MULTI-STEP, REAL-WORLD PROBLEMS WITH FRACTIONS OR DECIMALS – PART 2



LESSON INTRODUCTION

MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.

LEARNING TARGETS

- I can solve multi-step, real-world problems involving any of the four operations with decimals or fractions.

BENCHMARK COHERENCE

BUILDING ON

MA.5.AR.1 Solve problems involving the four operations with whole numbers and fractions.

WORKING TOWARD

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

ESSENTIAL QUESTIONS

- What different strategies can be used to solve multi-step equations with decimals and fractions?
- How can you tell what steps and operations are needed to solve multi-step problems?
- How do you know if your answer to a multi-step problem is reasonable?

LESSON NARRATIVE

In this lesson, students continue to solve real-world, multi-step word problems involving rational numbers. Students apply the problem-solving skills they developed in previous lessons from this unit to solve real-world problems. In the Collaboration, students specifically apply these skills to analyze the distance between hurdles in a 110-meter hurdle race, then determine which arithmetic operations are relevant and use them to solve the problems. Additionally, students draw or use a diagram to help them make sense of measurements, as well as to communicate their reasoning about the measurements. The Wrap-Up asks students to evaluate their understanding and comfortability with solving multi-step problems in real-world scenarios.

COMPONENT SUMMARY

3.9.1 WARM-UP (~5 MIN.)

Explore short division method

3.9.3 YOUR TURN! (20 MIN.)

Practice solving multi-step problems

3.9.2 COLLABORATION (15 MIN.)

Analyze and solve problems related to a 110-meter hurdle race

3.9.4 WRAP-UP (~5 MIN.)

Solve multi-step problems

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- If you plan to go to the school track, prepare hurdles or something that will mimic the hurdles and measuring tools.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may make computational errors when performing operations on multi-digit decimal and fraction problems.
- Students may have difficulty identifying which order of steps to take to find the correct answers.
- In 3.9.2 Collaboration, students may create 10 spaces (instead of 9) between 10 hurdles, leading to a miscalculation of the distance between hurdles and incorrect division in the following steps.

THINGS TO CONSIDER

- To foster flexible thinking and development of problem-solving skills, ask students to draw a visual model first and then create a numeric expression from the visual model.
- For the context of the 110-meter hurdles in 3.9.2 Collaboration, the distance between the first and last hurdles is 82.26 meters. If students end up with a non-terminating decimal, have them to revisit each step to find the error.

LITERACY AND LANGUAGE ROUTINES

Clarify, Correct, Critique

In 3.9.1 Warm-Up students analyze a new method of division, short division. To strengthen metacognitive thinking and problem-solving skills, consider positioning students to clarify, correct, and/or critique their own and others' reasoning by analyzing the new method, and comparing it to the standard long division algorithm.

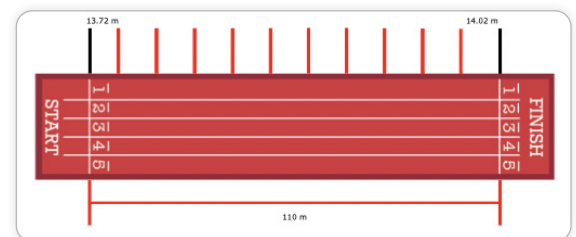
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

In 3.9.3 Your Turn!, students use problem-solving structures developed in lessons throughout this unit to solve multi-step, real-world problems. Students recognize relevant details, develop a logical sequence of steps, solve the problems, and check their work. Consider explicitly pointing out the patterns and structures among problems to ensure students recognize the connections.

DIFFERENTIATE INSTRUCTION

Use a real-life track at your school to bring 3.9.2 Collaboration to life. The manipulative, life-sized learning from visiting an actual track will activate visual and kinesthetic entry-points. 3.9.3 Your Turn! includes eight real-world problems with decimals and fractions. Consider structuring this component using a variety of methods according to student needs – this could include teacher-led small groups, a jigsaw activity, student choice, a relay race, circuit, or review game.



3.9.1 WARM-UP: SHORT DIVISION

Component Overview: Students investigate another method of division, short division.

TEACHER MOVES

Clarify, Correct, Critique

Students analyze a new method of division, short division. To strengthen metacognitive thinking and problem-solving skills, consider positioning students to clarify, correct, and/or critique their own and others' reasoning by analyzing the new method, and comparing it to the standard long division algorithm.



3.9.1 Warm-Up: Short Division

In Mexico, when dividing by a single-digit divisor, sometimes a method called "short division" is used. Consider the example shown of $74 \div 4$.

$$\begin{array}{r} 18 \\ 4 \overline{)74} \end{array}$$

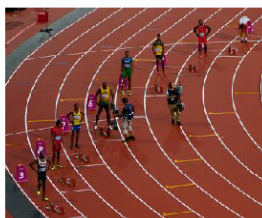
The quotient 18 is written above the dividend 74. The 1 is above the 7, the 8 is above the 4. A horizontal line is drawn under the 74. The divisor 4 is to the left of the dividend. The remainder 2 is written below the 4.

- Using your knowledge of division, explain where you think the 1, 3, 8, and 2 came from in the process.
First, the tens place is divided. 40 divides into 70 only 1 time so that is where the 1 comes from. 3 represents the remainder 30. Next the ones place is divided. 4 goes into 34 times and has a remainder of 2.
- Show how you might represent the same division problem using a method different from "short division."
Answers will vary.



3.9.2 Collaboration: Distance Between Two Hurdles

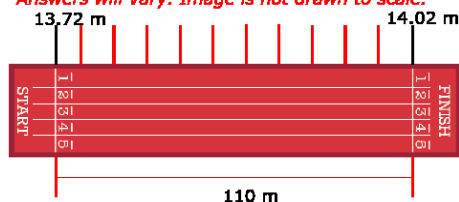
There are 10 equally spaced hurdles on a racetrack. The first hurdle is 13.72 meters from the start line. The final hurdle is 14.02 meters from the finish line. The racetrack is 110 meters long.



The diagram below represents what the track would look like if the track were straight and not curved.

- Use the diagram below to show the placement of the hurdles on the racetrack. Label all known measurements.

Answers will vary. Image is not drawn to scale.



- Discuss with your partner how many spaces there are between the 10 hurdles.

9 spaces

- How far are the hurdles from one another? Explain or show your reasoning.

$$110 - 13.72 - 14.02 = 82.26$$

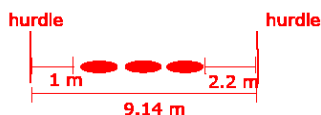
$$82.26 \div 9 = 9.14$$

9.14 meters

- A professional runner takes 3 strides between each pair of hurdles. The runner leaves the ground 2.2 meters before the hurdle and returns to the ground 1 meter after the hurdle.

- Draw a diagram to model the runner's strides between the hurdles. Label all known measurements.

Answers will vary. Image not drawn to scale.



- About how long are each of the runner's strides between the hurdles? Show your reasoning.

$$9.14 - 1 - 2.2 = 5.94$$

$$5.94 \div 3 = 1.98$$

1.98 meters

3.9.2 COLLABORATION: DISTANCE BETWEEN TWO HURDLES

Component Overview: Students collaborate to analyze and answer questions about the distance between hurdles in a 110-meter hurdle race. Prior to starting the activity, consider showing a video and/or images of 110-meter hurdle races and explain to students that the starting blocks are placed so that the outer and inner lanes still have the same running distance. Consider placing students in small groups based on learning styles and encourage visual and kinesthetic learners to act out or create a model of the scenario. Students then use these models to make sense of the measurements found and communicate their reasoning about the measurements.

TEACHER MOVES

Before students begin, consider discussing how the hurdles are staggered to compensate for the inner lanes being a shorter distance than the outer lanes. Also, discuss why there are 9 spaces between the 10 hurdles.

DIFFERENTIATE INSTRUCTION

Do you have track student-athletes in our class, or do you have a track at the school? Consider taking a field trip and implementing this lesson on your school's track outside to activate visual and kinesthetic entry-points.

ENRICHMENT

For questions 2 and 3, if students make arithmetic errors when calculating the distance between the first and last hurdles (82.26 meters), ask students to check their work with questions like:

- How many spaces are needed between 2 hurdles? 3 hurdles? 4 hurdles? How can you extend the pattern to 10 hurdles?
- Is your answer reasonable? How can you use estimation and rounding to check?

TEACHER MOVES

Clarify, Correct, Critique

As collaborating groups finish and have a completed diagram for question 4a and answer for question 4b, consider combining groups and positioning students to clarify, correct, and/or critique the responses of others. Groups can take turns sharing their diagrams, then the other students can ask clarifying questions, offer suggestions for correction, and critique the diagram for accuracy.

3.9.3 YOUR TURN!

Component Overview: Students practice solving multi-step problems with fractions and decimals. Consider using some or all problems based on learner needs for additional practice. Alternately, consider using certain problems in a teacher-led small group focused on helping students who may need additional support.

TEACHER MOVES

If students are working in partners, encourage identification and sharing of strategies and solution statements they are using for each question. By having students talk about their strategies and recognize the numeric expressions and operations used to solve, students build metacognitive thinking when problem solving. If time allows, partners can then share their answers with others to compare, contrast, and critique methods and solutions.

ENRICHMENT

To build problem-solving skills and connect solving two-step to three-step problems, ask scaffolding questions approaches like:

- What is the question asking you?
- What is the first step and what operation are you going to use?
- What is the second step and what operation are you going to use?
- What is the final step and operation needed to solve?

DIFFERENTIATE INSTRUCTION

Encourage students to draw visual models prior to attempting to solve the question. Make connections between the pictorial representation and the numeric expression explicit.



3.9.3 Your Turn!

1. Carpeting costs \$9.99 per yard. If Ginger buys 18.6 yards, how much will it cost her if she has a \$10 off coupon? **\$175.81**
2. A runner jogs $7\frac{1}{5}$ miles east, $5\frac{1}{4}$ miles south, and $8\frac{2}{3}$ miles west. What is the total number of miles jogged overall? **$21\frac{7}{60}$ miles**
3. Theta drinks $11\frac{1}{5}$ ounces of water at breakfast, $19\frac{1}{2}$ ounces at lunch, and $20\frac{2}{7}$ ounces at dinner. How many additional ounces will she need to drink in order to meet her 64-ounce goal? **$13\frac{1}{70}$ ounces**
4. A rectangular garden has a length of 8.75 inches and a width of 2.4 inches. If one bag of soil covers 6 square inches, how many bags would be needed to fill the garden? **4 bags**
5. Lawrence paid \$11.20 for 4 gallons of gas and an additional \$0.78 in tax. How much was the cost for each gallon of gas including tax? **\$3.00**
6. Frannie purchased a new television for \$799.99 with a sales tax rate of 7% of the purchase. If she splits the payments into equal amounts over 6 months, how much would she need to pay each month? **\$142.66**
7. A local farmer is constructing a new pen for his pigs. He has $608\frac{15}{16}$ feet worth of fencing material. Three of the four sides measure $89\frac{1}{4}$ feet, $147\frac{3}{8}$ feet, and $162\frac{1}{2}$ feet. How much fencing material remains after his 3 sides are installed? **$209\frac{13}{16}$ feet**
8. Rhonda bought 18 balloons priced at \$0.75 per balloon and 18 noisemakers priced at \$1.39 each. How much did Rhonda spend on her total purchase? **\$38.52**



3.9.4 Wrap-Up: Ticket Out the Door

1. Todd Everett's Flooring in Okeechobee, Florida sells a variety of flooring options. The company pays \$2.562 per square foot for specialty vinyl plank. The general manager purchases 4400 square feet of the vinyl plank to have in stock inventory. If the general manager has \$17,500.00 budgeted for stock inventory purchases, how much money does he have left in his budget for additional stock inventory after purchasing the specialty vinyl plank?
\$6,227.20
2. Tula is transitioning her 10-month old puppy, Rose, to adult dog food. To transition her puppy, Tula mixes adult and puppy dog food. Tula needs $\frac{9}{24}$ cup of adult food for Rose and $1\frac{1}{4}$ cups of adult food for her other dog, Ash. Tula only has a $\frac{1}{8}$ cup scoop. How many scoops of adult dog food will Tula need to have enough to feed Rose and Ash?
13 scoops of adult dog food

3.9.4 WRAP-UP: TICKET OUT THE DOOR

Component Overview: Consider using this as a formative assessment on the unit's focus on solving multi-step, real-world problems involving decimals and fractions. Students should not use a calculator to complete the Wrap-Up.

DIFFERENTIATE INSTRUCTION

Consider utilizing the online Check Your Understanding, EdgeXL, and Test Yourself! tools as additional practice and assessment opportunities for students.